

DIGITAL SAT MATH PRACTICE

Radian and Degree Conversion

15 Questions | ~15% Grid Ins | Difficulty Labeled

TOPIC: RADIAN – DEGREE CONVERSION

Question 1

[Easy]

An angle measures 150 degrees. What is the measure of this angle in radians?

- A) $2\pi/3$
- B) $5\pi/6$
- C) $3\pi/4$
- D) $7\pi/6$

Question 2

[Easy]

An angle has a measure of $3\pi/4$ radians. What is the degree measure of this angle?

- A) 120
- B) 135
- C) 150
- D) 225

Question 3

[Easy]

Which of the following is equivalent to 270 degrees in radians?

- A) $\pi/2$
- B) π
- C) $3\pi/2$
- D) 2π

Question 4**[Medium]**

A gear rotates through an angle of $7\pi/5$ radians. What is the degree measure of this rotation?

- A) 252
- B) 248
- C) 256
- D) 244

Question 5**[Medium]**

A wheel completes 2.5 full rotations. How many radians has it turned through?

- A) 4π
- B) 5π
- C) 2.5π
- D) 10π

Question 6**[Medium]**

The minute hand of a clock sweeps through 4 hours. What is the angle, in radians, that the minute hand has swept?

Student Produced Response (Grid In)

Question 7**[Medium]**

If an angle θ in standard position has its terminal side in Quadrant III and the reference angle is $\pi/6$, what is the value of θ in degrees?

- A) 150
- B) 210
- C) 240
- D) 330

Question 8**[Medium]**

A circular arc has a central angle of $2\pi/9$ radians. What is the degree measure of this central angle, rounded to the nearest whole number?

Student Produced Response (Grid In)

Question 9**[Medium]**

An arc on a circle of radius 12 cm subtends a central angle of 225° . What is the central angle in radians?

- A) $3\pi/4$
 - B) $5\pi/4$
 - C) $7\pi/4$
 - D) $5\pi/3$
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Question 10**[Hard]**

An angle measures $11\pi/6$ radians. What is the degree measure of this angle?

- A) 300
 - B) 315
 - C) 324
 - D) 330
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Question 11**[Hard]**

A pendulum swings through an angle of 0.35 radians on each side of vertical. What is the total angle of the swing, in degrees, rounded to the nearest tenth?

Student Produced Response (Grid In)

Question 12**[Hard]**

A satellite orbits Earth and sweeps an angle of $11\pi/12$ radians around the center of the Earth. A second satellite sweeps an angle of 168° . Which satellite has swept through a greater angle, and by how many degrees?

- A) The first satellite, by 3°
- B) The second satellite, by 3°
- C) The first satellite, by 15°
- D) They swept through equal angles

Question 13**[Hard]**

An engineer models the position of a piston with the function $P(t) = 8\cos(\pi t/3)$, where t is in seconds and P is in centimeters. The argument of cosine is in radians. What is the argument of cosine, in degrees, when $t = 5$?

- A) 60
 - B) 150
 - C) 240
 - D) 300
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Question 14**[Hard]**

Two angles are supplementary. One angle measures $(3k + 15)^\circ$ and the other measures $(k\pi/6)$ radians. What is the value of k ?

- A) 3
 - B) 5
 - C) 10
 - D) 15
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Question 15**[Hard]**

A Ferris wheel completes one full revolution every 90 seconds. After 25 seconds, through how many radians has a seat on the Ferris wheel rotated? (Give your answer as a fraction involving π .)

- A) $5\pi/9$
- B) $\pi/5$
- C) $2\pi/9$
- D) $5\pi/18$